

"Technical Specifications & Requirements"

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■ Table of Contents

- 1. Technical Overview
- 2. System Requirements
- 3. Hardware Requirements
- 4. Software Requirements
- 5. Network Requirements
- 6. API Requirements
- 7. Database Requirements
- 8. AI/ML Requirements
- 9. Security Requirements
- 10. Performance Requirements
- 11. Integration Requirements
- 12. Testing Requirements
- 13. Deployment Requirements
- 14. Maintenance Requirements

1. Technical Overview

IdeaDNA is a full-stack AI-powered web application built on modern cloud-native architecture. The platform leverages serverless functions, managed databases, and third-party AI APIs to deliver real-time idea generation, evolution, and reverse-engineering capabilities. This document specifies all technical requirements for development, deployment, and maintenance.

2. System Requirements

2.1 Functional System Requirements

ID	Requirement	Priority
SYS-001	System must support 10,000+ concurrent users	P0
SYS-002	System must handle 1000+ API requests per minute	P0
SYS-003	System must process AI generation requests within 15 seconds	P0
SYS-004	System must support real-time streaming for AI responses	P1
SYS-005	System must maintain 99.9% uptime	P0
SYS-006	System must auto-scale based on traffic	P1
SYS-007	System must support multi-tenant data isolation	P0
SYS-008	System must handle graceful degradation during AI API failures	P1

3. Hardware Requirements

3.1 Development Environment

ID	Requirement	Priority
DEV-HW-001	Development machine: 16GB RAM minimum, 32GB recommended	P0
DEV-HW-002	Development machine: 4-core CPU minimum, 8-core recommended	P0
DEV-HW-003	Development machine: 256GB SSD storage minimum	P0
DEV-HW-004	Development machine: Stable internet connection (10Mbps+)	P0

3.2 Production Environment (Cloud)

ID	Requirement	Priority
PROD-HW-001	Vercel Pro Plan: Serverless functions, Edge Network	P0
PROD-HW-002	Railway/Supabase: PostgreSQL with 2 vCPU, 4GB RAM	P0
PROD-HW-003	Redis: Upstash or Railway, 1GB memory	P1
PROD-HW-004	Pinecone: Starter pod for vector search	P1
PROD-HW-005	CDN: Vercel Edge Network for static assets	P1

4. Software Requirements

4.1 Development Stack

Category	Technology	Version	Purpose
Runtime	Node.js	20.x LTS	Server runtime
Framework	Next.js	14.x	React framework with App Router
Language	TypeScript	5.x	Type safety
Styling	Tailwind CSS	3.x	Utility-first CSS
State (Client)	Zustand	4.x	Client state management
State (Server)	React Query	5.x	Server state + caching
ORM	Prisma	5.x	Database ORM
Auth	Clerk	Latest	Authentication
AI Framework	LangChain	0.1.x	LLM orchestration
AI Model	OpenAI GPT-4	API	LLM inference
Database	PostgreSQL	15.x	Primary database
Cache	Redis	7.x	Session + caching
Vector DB	Pinecone	Latest	Semantic search
Animation	Framer Motion	11.x	UI animations
Charts	D3.js	7.x	Data visualization
Graph	React Flow	12.x	Evolution trees
Testing	Playwright	Latest	E2E testing
Linting	ESLint	8.x	Code quality
Formatting	Prettier	3.x	Code formatting

5. Network Requirements

ID	Requirement	Priority
NET-001	HTTPS enforced on all endpoints (TLS 1.3)	P0
NET-002	HSTS headers with max-age 31536000	P0
NET-003	CORS configured for whitelisted origins only	P0
NET-004	WebSocket connections via WSS (encrypted)	P1
NET-005	CDN for static assets (Vercel Edge)	P1
NET-006	DDoS protection via Cloudflare/Vercel	P1
NET-007	Rate limiting per IP and per user	P0
NET-008	API response compression (gzip/brotli)	P1

6. API Requirements

6.1 REST API Standards

ID	Requirement	Priority
API-001	All APIs follow RESTful conventions	P0
API-002	JSON request/response format	P0
API-003	Consistent error response structure: {error, message, code}	P0
API-004	Pagination for list endpoints (cursor-based)	P1
API-005	Versioning via URL path (/api/v1/...)	P1
API-006	Rate limit headers in all responses	P1
API-007	OpenAPI/Swagger documentation	P2
API-008	Request/response logging for debugging	P1

7. Database Requirements

ID	Requirement	Priority
DB-001	PostgreSQL 15+ with pgvector extension	P0
DB-002	Connection pooling via PgBouncer (max 20 connections)	P0
DB-003	Automated daily backups with 30-day retention	P0
DB-004	Read replicas for heavy analytical queries	P1
DB-005	Database migrations via Prisma Migrate	P0
DB-006	Indexing strategy for all query patterns	P0
DB-007	Soft deletes for data recovery	P1
DB-008	Audit logging for data changes	P2

8. AI/ML Requirements

ID	Requirement	Priority
AI-001	OpenAI GPT-4 API for text generation	P0
AI-002	OpenAI text-embedding-3-small for embeddings	P0
AI-003	LangChain for agent orchestration	P0
AI-004	Structured output parsing (JSON mode)	P0
AI-005	Temperature tuning per agent (0.2 - 0.8)	P0
AI-006	Streaming responses for real-time UX	P1
AI-007	Fallback to cached responses on API failure	P1
AI-008	Token usage tracking per user	P1
AI-009	Rate limiting: 5-500 requests/hour based on tier	P0
AI-010	Prompt versioning and A/B testing capability	P2

9. Security Requirements

ID	Requirement	Priority
SEC-001	OAuth 2.0 + JWT for authentication	P0
SEC-002	RBAC with User and Admin roles	P0
SEC-003	AES-256 encryption for sensitive data at rest	P0
SEC-004	TLS 1.3 for all data in transit	P0
SEC-005	API keys stored in environment variables only	P0
SEC-006	Input validation via Zod schemas	P0
SEC-007	SQL injection prevention via Prisma ORM	P0
SEC-008	XSS prevention via React escaping + DOMPurify	P0
SEC-009	CSRF protection via SameSite cookies	P0
SEC-010	Rate limiting per user and per IP	P0
SEC-011	Security headers (HSTS, CSP, X-Frame-Options)	P0
SEC-012	Dependency scanning (npm audit, Snyk)	P1

10. Performance Requirements

ID	Requirement	Priority
PERF-001	Page load time < 2 seconds (First Contentful Paint)	P0
PERF-002	Time to Interactive < 3 seconds	P0
PERF-003	AI generation response < 15 seconds (deep-dive)	P0
PERF-004	Initial idea cards < 3 seconds	P0
PERF-005	API response time < 200ms (cached)	P0
PERF-006	Database query time < 100ms	P0
PERF-007	Lighthouse Performance score > 90	P1
PERF-008	Lighthouse Accessibility score > 90	P1
PERF-009	Lighthouse Best Practices score > 90	P1
PERF-010	Lighthouse SEO score > 90	P2

11. Integration Requirements

ID	Requirement	Priority
INT-001	OpenAI API integration with error handling and retries	P0
INT-002	GitHub API integration for repo analysis	P0
INT-003	Clerk Auth integration with webhooks	P0
INT-004	Vercel deployment integration with preview URLs	P0
INT-005	Railway/Supabase database integration	P0
INT-006	Redis cache integration	P1
INT-007	Pinecone vector search integration	P1
INT-008	Sentry error tracking integration	P1
INT-009	Mixpanel/Amplitude analytics integration	P2
INT-010	Stripe billing integration (future)	P3

12. Testing Requirements

ID	Requirement	Priority
TEST-001	Unit tests for all utility functions (Jest)	P0
TEST-002	Integration tests for API endpoints	P0
TEST-003	E2E tests for critical user flows (Playwright)	P0
TEST-004	AI agent output validation tests	P1
TEST-005	Performance tests (Lighthouse CI)	P1
TEST-006	Accessibility tests (axe-core)	P1
TEST-007	Security tests (OWASP ZAP)	P1
TEST-008	Cross-browser testing (Chrome, Firefox, Safari, Edge)	P1
TEST-009	Mobile responsive testing (iOS, Android)	P1
TEST-010	Load testing (k6 or Artillery)	P2

13. Deployment Requirements

ID	Requirement	Priority
DEP-001	CI/CD pipeline via GitHub Actions	P0
DEP-002	Automated testing on every PR	P0
DEP-003	Preview deployments for every PR	P0
DEP-004	Production deployment on merge to main	P0
DEP-005	Database migrations run automatically	P0
DEP-006	Environment variables managed via Vercel dashboard	P0
DEP-007	Rollback capability within 5 minutes	P1
DEP-008	Blue-green deployment for zero downtime	P2

14. Maintenance Requirements

ID	Requirement	Priority
MAINT-001	Automated dependency updates (Dependabot)	P1
MAINT-002	Weekly security vulnerability scans	P0
MAINT-003	Monthly performance reviews	P1
MAINT-004	Quarterly database optimization	P2
MAINT-005	AI model performance monitoring	P1
MAINT-006	User feedback collection and analysis	P1
MAINT-007	Documentation updates with every release	P1
MAINT-008	Disaster recovery testing (bi-annual)	P2

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